



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

Name: Lorilla, Zhay S.	Date: 09-24-2026
Section: BSIT-2A	Instructor: Ms. Maxyne Nuela Ignacio

I. Objective

An IT Automation Incident Ticket Manager that tracks incidents generated by automation bots. Each ticket, with its Incident ID, Bot and Short Description, is stored in a list, which is a linear data structure. A menu lets the user add, display, search, remove and count tickets, using the 10 preloaded samples for testing.

II. Program Operations

Add Incident:

```
tickets.append({"id": i, "bot": b, "desc": d})
```

After checking for blank fields and duplicate IDs, this line adds the new ticket to the end of the list.

Display Incidents:

```
for t in tickets:  
    print(f"{t['id']:<15}{t['bot']:<20}{t['desc']}]")
```

This loop goes through the list in order and prints each ticket's ID, bot and description.

Search Incident:

```
for t in tickets:  
    if t["id"].upper() == i.upper():
```

This loop checks the tickets one by one and compares each ID with the one entered. When it finds a match, it prints the ticket's details.

Remove Resolved Incident:

```
tickets.remove(t)
```

Once the loop finds the ticket with the matching ID, this line deletes it from the list.

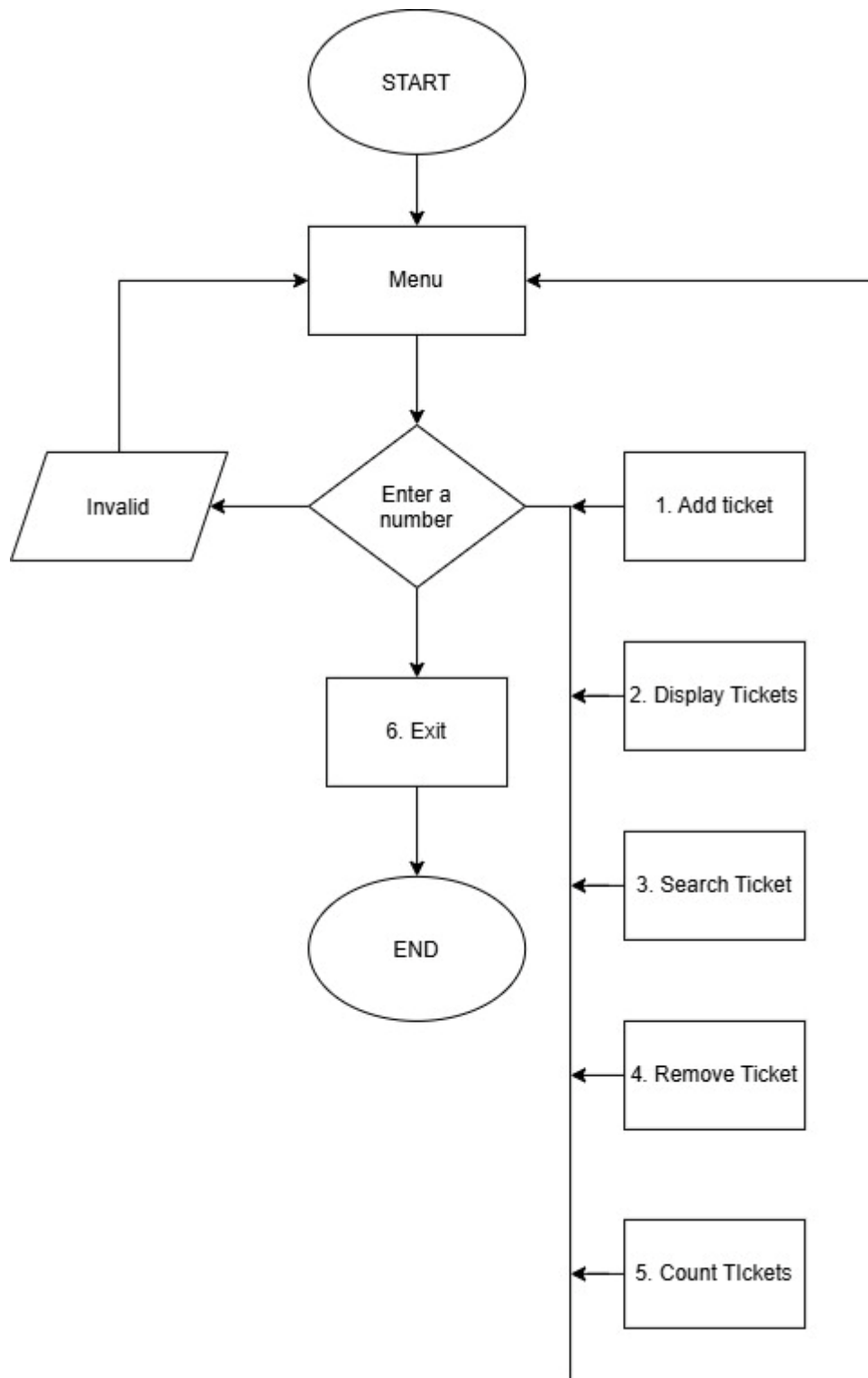
Count Active Incidents:

```
print("Total active tickets:", len(tickets))
```

len() returns the number of tickets left in the list, and this line prints it.



III. Flowchart



IV. Conclusion

I learned how to store and manage records with a linear data structure. Each ticket is kept in a list, where items are arranged one after another in sequence, so adding, displaying, searching, removing and counting all work by moving through that list.